Reg. No. | | | | | | | | | | |

[12EC7608]

SEVENTH SEMESTER B.E EXAMINATION DECEMBER - 2015

IMAGE PROCESSING

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	**a.**	With a neat block diagram, explain the fundamental steps in image processing.	**08**
	b.	Explain the following terms: i. Brightness adaptation ii. Weber ratio iii. Single imaging sensor	**06**
2.	**a.**	Explain the role of image sampling and quantization in the digital image processing.	**07**
	b.	Explain the following terms in image processing: i. Neighbors of a pixel ii. Distance measures	**07**
3.	**a.**	Compute Haar transform for N=4.	**08**
	b.	Explain any four properties of two dimensional Fourier transform.	**06**
4.	**a.**	With necessary graphs, explain the following spatial image enhancement operations: i. Image negative ii. Log transformation iii. Power law transformation iv. Contrast stretching	**08**
	b.	Explain the use of second order derivatives for image sharpening.	**06**
5.	**a.**	Explain the different smoothing filters used in frequency domain with relevant expressions and graph.	**07**
	b.	Explain homomorphic fittering with suitable derivation.	**07**
6.	**a.**	Explain the basic model of image restoration process also, with necessary equations, explain the most common PDFs in an image processing.	**08**

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	b.	With relevant mathematical equations, explain inverse fittering and geometric mean filter.	06
7.	a.	Discuss briefly any two color models used in color image processing.	07
	b.	Explain the procedure in converting colors from RGB to HIS and Vice Versa.	07
8.	a.	What is the need for image compression? Explain the compression model.	06
	b.	Explain the following basic compression method with suitable example: i. Huffman coding ii. Arithmetic coding	08